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Started	Wednesday, 21 October 2026, 9:53 PM
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Duration	4 mins 33 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It always starts from the beginning of the list
c. It compares elements sequentially
d. It can be applied only if the list is sorted ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list contains duplicate elements
c. When the list is very small
d. When the list is unsorted

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 89 and 99
c. 90 and 99 ✓
d. 90 and 94

Question **6**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The list is sorted
d. The search continues in the right sublist

Question **7**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Sort the list
c. Compare the target element with the middle element in the list ✓
d. Divide the list into two equal parts

Question **8**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Linear search
b. Hash search ✗
c. Binary search
d. Interpolation search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O👎
c. O(1)
d. O(n log n)

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 1000
c. 100
d. 10 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Sorting elements in a list
- c. Inserting elements into a list
- d. Determining whether an element is present in a list ✓

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. log n
- b. n ✓
- c. n/2
- d. n-1

Question **13**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. At the beginning of the list
- b. In the right sublist ✓
- c. In the middle of the list
- d. In the left sublist

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. Number values should only be present

Number values should only be present

d. There should be direct access to the middle element in any sublist

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element

b. Binary search does not require dividing the list

c. Binary search works on unsorted lists

d. Binary search has a lower time complexity on large, sorted lists ✓